

1 Title

Feature Selection for Optimizing Glioblastoma Treatment Decisions

2 Group Members

Roshan Mahesh

3 Abstract

Glioblastoma (GBM) is an aggressive brain cancer with limited treatment options. The standard treatment, Temozolomide (TMZ) and Radiation Therapy (RT), does not work equally well for all patients. Some patients respond well and go on to live longer lives, while others experience rapid disease progression despite receiving the same treatment. To further add to the area of personalized treatment plans, this project aims to predict which GBM patients will benefit most from TMZ and/or RT by leveraging single-cell immune and tumor markers. Using SHAP-based feature selection in conjunction with the Light Gradient Boosting Machine (LightGBM) algorithm, I aim to identify the most important biomarkers that correlate with treatment response, which could optimize treatment strategies for GBM patients by providing personalized treatment selection, avoiding ineffective treatments, and identifying treatment-resistant subtypes.

4 Introduction

4.1 Problem Motivation

GBM is the most common type of malignant brain tumor among adults, with more than 13,000 Americans receiving this diagnosis each year[1]. Known to be an extremely aggressive form of cancer, GBM patients have a median survival timeline of approximately 12 to 15 months, despite standard treatments such as TMZ and RT[1]. The two-year survival rate for patients receiving both TMZ and RT is about 26.5%, compared to 10.4% for those receiving RT alone[1]. Currently, there are not any reliable methods to predict individual outcomes as patient responses to these treatments vary significantly, with some patients being able to live upwards of 54 months while others experience quick decline and pass away within a few months[4]. While there are methods such as identifying predictive biomarkers, such as MGMT promoter methylation status, that have shown promise in select demographics in forecasting treatment responses, it's clear that further research in the area will enable personalized GBM therapy, reduced ineffective treatments, and guide alternative therapeutic strategies for patients that are flagged as non-responders to common treatments.

4.2 Previous work focused on solving this problem

When looking at the current literature, there has been significant growth in prioritizing individualized treatment plans, including with GBMs. Specifically, in a paper by Leelatian et al., titled "Unsupervised machine learning reveals risk stratifying glioblastoma tumor cells", the authors profiled the tumor-infiltrating immune and tumor cells in glioblastoma using single-cell mass cytometry across 28 patient samples. They applied t-SNE for dimensionality reduction and used FlowSOM clustering to organize the data into 43 distinct cellular clusters[3]. To assess the prognostic significance of each cluster, the authors calculated the hazard ratio of death and corresponding p-values based on overall patient survival. This analysis identified five Glioblastoma Positive Prognostic (GPP) clusters and four Glioblastoma Negative Prognostic (GNP) clusters[3]. The GNP clusters, which were associated with poorer survival outcomes, exhibited high expression of neural-lineage markers such as S100B and SOX2, along with activation of pro-survival signaling proteins including phosphorylated STAT3, STAT5, and S6[3]. In contrast, the GPP clusters, associated with improved survival,

displayed high expression of EGFR and CD44 but lacked activation of the pro-survival pathways observed in GNP clusters[3]. Rather than developing predictive models, the study focused on unsupervised stratification of cellular states and their association with clinical outcomes. Their results highlighted the prognostic relevance of tumor-infiltrating immune phenotypes and pointed to signaling programs that could serve as potential therapeutic targets. However, no modeling was performed to predict patient-level outcomes using these cellular features.

Another study by Wang et al. investigated the feasibility of detecting O⁶-methylguanine-DNA methyltransferase (MGMT) promoter methylation using minimally invasive liquid biopsy methods. MGMT promoter methylation is a well-established prognostic biomarker in gliomas, especially among older age groups, and is often associated with better response to alkylating chemotherapy[7]. The study analyzed samples from 89 glioma patients and compared MGMT promoter methylation status in tumor tissue, serum, and cerebrospinal fluid (CSF) using methylation-specific polymerase chain reaction (PCR). MGMT promoter methylation was detected in 57.3% of tumor tissues[7]. Among liquid biopsies, CSF exhibited a higher sensitivity (65.0%) than serum (37.3%), with both demonstrating 100% specificity relative to tumor tissue[7]. These findings highlight the potential of CSF as a more reliable non-invasive medium for detecting MGMT promoter methylation in GBM patients. Importantly, CSF-based detection remained robust across glioma grades and showed enhanced sensitivity in patients with residual postoperative tumors[7]. This suggests a promising clinical application of CSF liquid biopsies in diagnosis and prognosis, particularly in cases where tissue sampling is challenging. The authors emphasize the need for larger prospective studies to validate these results and support the integration of CSF-based methylation analysis into routine clinical workflows for glioma management.

4.3 Limitations of previous work

Although previous approaches have provided important insights into glioblastoma biology, they fall short in delivering clinically actionable, patient-specific predictions. For instance, Leelatian et al. employed unsupervised clustering and hazard modeling to define prognostic cellular states based on single-cell mass cytometry, revealing GNP and GPP clusters associated with patient survival[3]. However, their method lacked a predictive framework and could not generalize findings to new patients or quantitatively stratify individuals by treatment response. Similarly, Wang et al. demonstrated the feasibility of detecting MGMT promoter methylation through liquid biopsies, particularly in CSF, but this approach was limited to a single binary biomarker and did not leverage the full heterogeneity of cellular phenotypes or integrate multi-marker signals into a predictive model.

Other computational methods, such as Citrus and Cytofast, also face key limitations when applied to glioblastoma. These models typically require supervision using categorical outcomes, which are not always clearly defined or available, especially when dealing with continuous clinical variables like progression-free survival or tumor burden[5]. Moreover, given the profound intratumoral heterogeneity of GBM at both the genetic and molecular levels, a single biopsy often fails to capture the full complexity of the tumor microenvironment[6]. This makes it difficult for marker-driven clustering models to produce consistent or interpretable stratifications, especially when multiple subpopulations may express overlapping or context-dependent markers. In contrast, our approach combines engineered features from millions of single-cell measurements with interpretable, cross-validated machine learning to deliver individualized, outcome-driven predictions that account for cellular heterogeneity and support clinical decision-making.

5 Statement of Contributions

In this paper, I developed a machine learning framework for predicting GBM patient treatment response using single-cell mass cytometry data. The contributions of this work can be summarized as follows. First, I introduced a biomarker selection pipeline that combines SHAP-based feature importance estimation with LightGBM models. Second, I engineered a comprehensive set of per-patient features, including marker mean, standard deviation, and percent positive expression percentage to capture cellular heterogeneity more robustly than traditional marker summaries. Third, I demonstrated the predictive application of these engineered features by achieving high cross-validated and holdout-set accuracy in classifying patients as responders or non-responders, which provides a potential computational foundation for personalized treatment strategies in GBM. Finally, my approach moves beyond prior unsupervised analyses of GBM mass cytometry data by directly modeling clinical outcomes, which bridges cellular-level measurements and patient-level therapeutic decisions..

6 Methods

6.1 Notation

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$ denote the dataset, where each $x_i \in \mathbb{R}^d$ represents a per-patient feature vector constructed from the following aggregated single-cell mass cytometry measurements: mean, standard deviation, and percent positive expression. The corresponding label $y_i \in \{0, 1\}$ indicates treatment response, defined by overall survival time as patients with survival duration greater than the cohort median were labeled as responders ($y_i = 1$), and those below the median as non-responders ($y_i = 0$). The training set is denoted $\mathcal{D}_{\text{train}} \subset \mathcal{D}$ and the disjoint holdout test set as $\mathcal{D}_{\text{test}} = \mathcal{D} \setminus \mathcal{D}_{\text{train}}$. Each LightGBM model trained during cross-validation is denoted f_k , the final model is f_{final} , Shapley values are denoted $\phi_j^{(k)}(x)$, and aggregated feature importances are denoted I_j .

6.2 Problem Formulation

The objective is to learn a predictive function $f : \mathbb{R}^d \rightarrow \{0, 1\}$ that maps patient-level features to binary treatment response outcomes. To select stable predictive biomarkers, stratified K -fold cross-validation is performed on $\mathcal{D}_{\text{train}}$, partitioning it into K disjoint subsets $\{\mathcal{D}_1, \dots, \mathcal{D}_K\}$. For each fold k , a LightGBM model f_k is trained on $\mathcal{D}_{\text{train}} \setminus \mathcal{D}_k$, where LightGBM models the prediction as an additive ensemble of decision trees:

$$\hat{y}(x) = \sum_{m=1}^M f_m(x),$$

by minimizing a regularized first- and second-order Taylor approximation of the logistic loss function. Shapley values $\phi_j^{(k)}(x)$ are computed to measure the contribution of each feature j to model predictions, and feature importance is aggregated across folds as:

$$I_j = \frac{1}{K} \sum_{k=1}^K |\phi_j^{(k)}|.$$

The top M features with the highest average absolute SHAP values are selected. A final LightGBM model f_{final} is then trained on $\mathcal{D}_{\text{train}}$ restricted to these selected features, and evaluated on the independent holdout test set $\mathcal{D}_{\text{test}}$.

6.3 Description of your method

The input to the model consists of engineered feature vectors x_i , where each vector aggregates the single-cell mass cytometry measurements collected for a given patient, by computing the mean, standard deviation, and

percent positive expression for each of the 43 biomarkers. Across the cohort, approximately two million single-cell measurements were collected from 28 patients in total. I computed the mean, standard deviation, and percent positive expression to capture the central tendency, cellular heterogeneity, and prevalence of marker expression, respectively, enabling a comprehensive characterization of tumor biology at the patient level. The binary output $y_i \in \{0, 1\}$ indicates treatment response status, determined by whether a patient's overall survival time exceeded the cohort median. To prevent information leakage, the dataset $\mathcal{D}$ was partitioned into an 65% training set $\mathcal{D}_{\text{train}}$ and a 35% holdout test set $\mathcal{D}_{\text{test}}$ prior to any feature selection or model training. Within $\mathcal{D}_{\text{train}}$, stratified 5-fold cross-validation was conducted to ensure that feature selection was based only on training data and to reduce variance in feature importance estimates, as 5-fold CV provides a good balance between bias and variance given the limited patient sample size. For feature selection, Shapley values were computed on each cross-validation fold because they provide a theoretically grounded measure of each feature's marginal contribution to model predictions, allowing robust identification of biologically meaningful biomarkers even in the presence of feature interactions. The top M features with the highest aggregated SHAP importance scores across folds were selected. A final LightGBM model f_{final} was trained exclusively on these selected features and evaluated on the unseen test set $\mathcal{D}_{\text{test}}$ to assess generalization performance. LightGBM was selected due to its ability to model nonlinear feature interactions, handle small noisy datasets effectively due to the effects of boosting, and provide interpretable feature attributions through SHAP analysis.

6.4 Schematic illustration of your method

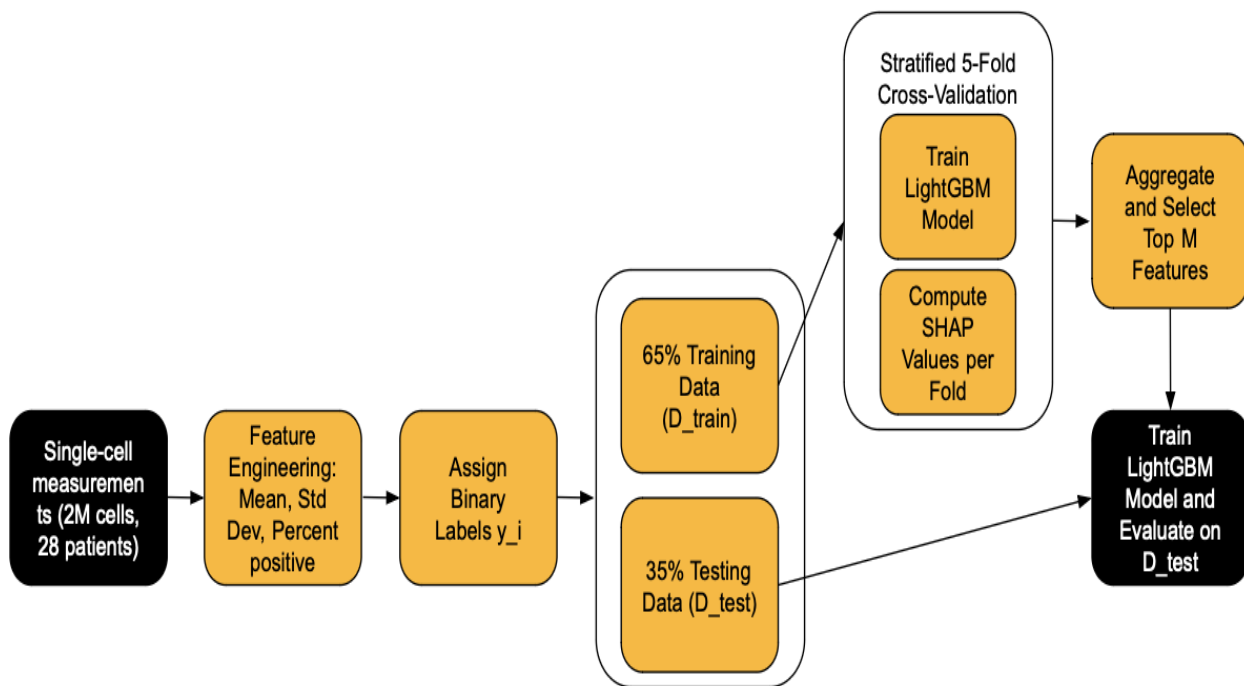


Figure 1: Architecture of Project

As you can see in Figure 1, the entire process that I have discussed in earlier sections is now illustrated. The input and output are clearly shown with a black background box, and the division between training and testing data is shown to distinguish the data split that was used to prevent data leakage and inconsistencies in the results.

7 Results

Table 1: Confusion Matrix

	Predicted Non-Responder	Predicted Responder
True Non-Responder	3	2
True Responder	0	5

Table 2: Performance Metrics

Metric	Non-Responder (0)	Responder (1)
Precision	1.00	0.71
Recall	0.60	1.00
F1-score	0.75	0.83
Overall Accuracy	0.80	

Table 3: Top 10 Features Selected in Each Fold and Most Frequent Features Across Folds

Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Most Frequent
pAKT-152_pct_pos	Dead-103_std	MET-162_std	pAKT-152_std	CD117-143_std	MET-162_std
CD15-164_mean	cCasp3-142_pct_pos	gH2AX-147_pct_pos	CD117-143_mean	CD15-164_std	Dead-103_pct_pos
CD44-169_std	MET-162_std	SOX2-163_pct_pos	CycinB1-139_std	Dead-103_pct_pos	CD15-164_mean
pSTAT3-158_pct_pos	NA-191_mean	Dead-103_mean	MET-162_mean	CycinB1-139_std	EGFR-165_mean
HLADR-174_std	CD15-164_mean	CycinB1-139_mean	CD31-145_pct_pos	CycinB1-139_mean	cCasp3-142_pct_pos
HH3-176_std	TUJ1-141_pct_pos	EGFR-165_mean	MET-162_std	CD117-143_mean	Dead-103_mean
MET-162_std	140_mean	MET-162_pct_pos	CD56-155_pct_pos	CD34-148_pct_pos	CD117-143_std
149_mean	Dead-103_pct_pos	pERK-171_pct_pos	cCasp3-142_pct_pos	NA-191_std	CycinB1-139_mean
EGFR-165_mean	Dead-103_mean	Dead-103_pct_pos	pHH3-175_pct_pos	pSTAT3-158_mean	CD117-143_mean
pSTAT3-158_std	CD117-143_std	CD56-155_mean	S100B-144_std	140_pct_pos	CycinB1-139_std

7.1 Datasets

The dataset utilized in this study consists of approximately two million single-cell mass cytometry measurements collected over an extended period from 28 GBM patients. For each patient, mass cytometry was performed to quantify the expression levels of 43 biomarkers, resulting in high-dimensional, cell-level data. Each row in the raw dataset corresponds to a single cell, and each column corresponds to a normalized biomarker intensity measurement. To construct patient-level feature vectors suitable for downstream modeling, the mean, standard deviation, and percent positive expression were computed for each biomarker across all cells from a given patient, capturing central tendency, intra-patient heterogeneity, and population prevalence of marker expression, respectively. In addition to the mass cytometry data, patient-level clinical metadata was available, including gender, age, extent of surgical resection (gross total or subtotal), receipt of TMZ chemotherapy, receipt of RT, MGMT promoter methylation status, progression-free survival (PFS), and overall survival (OS) times. The metadata was primarily used to define treatment response labels:

patients with overall survival greater than the cohort median were categorized as responders, and those with survival below the median were categorized as non-responders. Together, the integrated single-cell and clinical datasets provide a comprehensive view of tumor cellular phenotypes and patient outcomes, enabling the development of predictive models for GBM treatment response.

7.2 Baselines

To further evaluate the biological relevance of the features selected by my supervised learning approach, I compared our results against the glioblastoma prognostic clusters identified by Leelatian et al. Their study defined GNP clusters by high expression of neural-lineage proteins such as S100B and SOX2, alongside activation of pro-survival signaling pathways including phosphorylated STAT3, STAT5, and S6, which, despite enhancing tumor cell survival, were associated with poorer overall patient survival outcomes[3]. Conversely, GPP clusters were characterized by elevated EGFR and CD44 expression but lacked activation of the pro-survival signaling programs typical of GNP clusters, and were linked with improved patient survival[3].

Consistent with these findings, several of the most frequently selected features in our model included EGFR-165_mean, a biomarker associated with GPP clusters and favorable prognosis, as seen in Table 3. Additionally, phosphorylated STAT3 (pSTAT3-158_pct_pos and pSTAT3-158_std), CD44-169_std, and S100B-144_std were repeatedly selected across folds, reflecting the activation of tumor cell pro-survival signaling characteristic of GNP clusters. These overlaps suggest that our LightGBM and SHAP-based feature selection framework was able to independently recover biologically meaningful patterns associated with survival outcomes, while additionally producing a predictive model capable of quantifying patient-specific treatment response likelihoods.

However, it was not possible to directly compare our model’s quantitative performance metrics to the baseline results from Leelatian et al. due to fundamental differences in dataset structure, study goals, and methodological frameworks, as their work focused on unsupervised clustering and biological characterization rather than outcome prediction.

7.3 Description of Experiments

To evaluate the predictive performance of the model, I partitioned the dataset into a 65% training set and a 35% holdout test set. SHAP-based feature selection and LightGBM model training were performed using stratified 5-fold cross-validation within the training set, and the final model was evaluated exclusively on the holdout set. Model performance was assessed using accuracy, precision, recall, and F1-score, as can be seen in Table 2. These metrics were selected because they jointly provide a rigorous assessment of predictive performance in a binary classification setting, particularly in clinical contexts where both correct identification of responders and non-responders is critical, which is also why I included a Confusion Matrix table as seen in Table 1. Accuracy alone would not be sufficient if the classes were imbalanced or if misclassification costs differed between groups. Therefore, precision and recall complement accuracy by highlighting how well the model captures true clinical outcomes while minimizing false positives and false negatives. The 80% accuracy observed, combined with strong precision and recall scores, suggests that the model generalizes well to unseen data without evidence of overfitting or data leakage.

In terms of feature interpretation, the top 10 most frequently selected biomarkers across folds included MET-162, Dead-103, CD15-164, EGFR-165, cCasp3-142, CD117-143, and CycinB1-139 markers, as seen in Table 3. Several trends emerged when comparing the features selected in each fold to the globally most stable features. Biomarkers related to tumor proliferation (e.g., CycinB1-139_mean and CycinB1-139_std), apoptotic signaling (e.g., Dead-103_mean, Dead-103_pct_pos, cCasp3-142_pct_pos), and oncogenic signaling (e.g., EGFR-165_mean, MET-162_std) were consistently ranked highly across folds, indicating that these

biological processes are robustly associated with treatment outcomes in glioblastoma. Immune markers such as CD15-164_mean and CD117-143_std also appeared frequently, suggesting a potential role of tumor-immune interactions in therapeutic response. Although there was some fold-to-fold variability in additional features (such as pAKT-152, S100B-144, and pSTAT3-158), the core set of top biomarkers remained highly stable across cross-validation folds, reinforcing their biological relevance. Together, these results suggest that features capturing tumor cell proliferation, survival signaling, and immune phenotypes are critical for predicting patient outcomes and could serve as potential biomarkers for stratifying glioblastoma patients in clinical settings.

8 Discussion

8.1 Recap

The goal of this work was to develop a machine learning model capable of predicting glioblastoma treatment response using patient-level features derived from single-cell mass cytometry data. To accomplish this, I aggregated per-cell biomarker measurements into engineered features—specifically, the mean, standard deviation, and percent positive expression for each of 43 markers—capturing central tendencies, variability, and cellular prevalence. Using these features, I trained a LightGBM classifier and applied SHAP-based feature selection within a strict cross-validation pipeline to identify the most informative biomarkers. This model achieved strong predictive performance on a held-out test set, with an overall accuracy of 80%, while also identifying biologically meaningful features involved in cell proliferation, apoptosis, and immune signaling. These results demonstrate that high-dimensional single-cell data, when properly aggregated and interpreted, can support clinically relevant predictions and reveal mechanistic insights into glioblastoma treatment outcomes.

8.2 Observations

This method enabled us to identify a compact and interpretable set of biomarkers that are strongly associated with treatment response in glioblastoma patients. By applying SHAP-based feature selection within a cross-validated LightGBM framework, I was able to uncover consistent biological signals related to tumor proliferation (e.g., MET, EGFR, CycinB1), apoptosis (e.g., cCasp3, Dead-103), and immune activity (e.g., CD15, CD117) that differentiated responders from non-responders. Compared to the baseline analysis performed in Leelatian et al., which used unsupervised clustering to define prognostic groups without predicting clinical outcomes, our supervised model was able to make quantitative, patient-level treatment predictions with 80% accuracy. Moreover, the top features strongly aligned with markers present in the GNP and GPP clusters described in that study, confirming that this predictive approach not only generalizes well but also independently rediscovers biologically relevant patterns associated with patient survival.

8.3 Limitations and Future Work

One of the primary limitations of this study is the small sample size: only 28 patients were available for analysis. While cross-validation and holdout evaluation helped mitigate overfitting, such a limited cohort restricts the generalizability of the findings and makes it difficult to draw statistically robust conclusions. Additionally, while the model identified biologically meaningful markers and achieved promising accuracy, its performance may vary significantly on independent datasets due to potential cohort-specific biases. The model also did not incorporate temporal dynamics or longitudinal measurements, which could provide deeper insight into treatment response trajectories. If more time and resources were available, the next steps would involve validating the model on a larger, multi-institutional dataset and integrating additional clinical and genomic features to improve predictive power and biological interpretability. Expanding the cohort size would also allow for more reliable biomarker discovery and more nuanced stratification of patient subgroups.

8.4 Inspiring Concluding Paragraph

This work demonstrates how machine learning, when thoughtfully applied to complex biological data, can reveal powerful insights into disease mechanisms and treatment outcomes. By linking single-cell biomarker patterns to patient survival in glioblastoma, one of the most aggressive and deadly cancers, this approach moves beyond descriptive studies to actionable prediction. It shows that with the right computational tools, the heterogeneity within tumors—which has long challenged effective treatment—can instead become a source of predictive strength. Although further validation is needed, this framework lays the foundation for a future where personalized treatment strategies are informed directly by the biological realities of each patient’s tumor. Ultimately, it represents a step toward more precise, data-driven medicine, where we use every available cell-level signal to better understand, predict, and overcome disease.

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